

Selection bias correction for imbalanced samples

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Abstract

Selection bias correction is generally crucial to have a reliable inference for a nonprobability sample. Often the correction is applied with the two-sample setup. That is, along with the nonprobability sample we are interested in, a probability sample sharing some common auxiliary variables is used for constructing correction weights for the nonprobability sample. The two-sample setup allows one to calculate weighted estimates for population parameters of interest based on the nonprobability sample. Since the nonprobability sample is usually easy to collect, we often end up with a large nonprobability sample and a small probability sample. The imbalance of the two samples may cause difficulties in modelling the propensity of units to be included in the nonprobability sample. This paper discusses some often-seen solutions for imbalanced samples in machine learning literature, i.e. undersampling, Synthetic Minority Oversampling Technique (SMOTE), and a mixture of both. A selection bias correction framework is adjusted to incorporate the imbalance solutions. Three evaluation studies across different types of data sets are shown. The results indicate that SMOTE has the potential to deal with imbalances in selection bias correction, while further study is needed to identify in which scenarios SMOTE works well.

Keywords descriptive inference, imbalanced sample, nonprobability sample, SMOTE, undersampling

1 Introduction

Many new data types emerge because of digitalization. Think of social media data, sensor data, or administrative data. These data are usually not based on a known sampling design and are termed nonprobability samples. If a nonprobability sample is treated as a simple random sample, selection bias may occur for population parameter estimation. To correct for selection bias, many correction methods adopt the two-sample setup. That is, along with the nonprobability sample, a probability sample sharing some common auxiliary variables from the same population is used, see for example, Elliott and Valliant (2017), Chen et al. (2020), and Liu et al. (2022). In the two-sample setup, by

Received: September 23, 2024. Revised: May 19, 2026. Accepted: May 20, 2026

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combining the nonprobability sample and the probability sample, the inclusion probabilities (or propensities) of the units in the nonprobability sample are estimated, and the reciprocals of these inclusion propensities can be used as weights for the units in the nonprobability sample. A design-based estimator (e.g. Hájek estimator) can then be applied for population parameter estimation. Other correction methods can also be found in recent reviews of selection bias correction methods, see for example, [Rao \(2021\)](#) and [Wu \(2022\)](#).

Given that a nonprobability sample is rather time and cost-efficient to collect, researchers often have a massive nonprobability sample and a relatively small probability sample. This means that researchers will end up with an imbalanced combined sample, which may cause difficulties when estimating the inclusion propensities of the nonprobability sample. The problem of imbalanced samples has been well-studied in many contexts. For example, [Wallace and Dahabreh \(2014\)](#) pointed out that the probabilities of the minority group are generally underestimated in an imbalanced sample. A rich discussion regarding the imbalance in the prediction task can also be found in, for example, [Estabrooks et al. \(2004\)](#) and [Batista et al. \(2004\)](#). However, to our best knowledge, the imbalance of the combined two-sample in selection bias correction has not been studied.

One straightforward approach to deal with imbalanced samples is to downsize the majority group. In our case, this would mean drawing a subsample of the nonprobability sample so that it has a similar size as the probability sample. A large proportion of the nonprobability sample may then be discarded from the propensity modelling process. Another popular way to deal with an imbalanced sample is the Synthetic Minority Oversampling Technique (SMOTE) proposed by [Chawla et al. \(2002\)](#). SMOTE creates synthetic samples for the minority group in a nonparametric manner. That is, weighted combinations of a unit and its neighbours are used to create synthetic ‘new’ samples. It has been shown that if the generated new samples follow the population distribution, SMOTE may considerably improve the prediction performance ([Elreedy & Atiya, 2019](#)). Many variants of SMOTE have also been proposed for different types of data or different goals of modelling. For example, SMOTE-ENC is for a data set containing both continuous and categorical data ([Mukherjee & Khushi, 2021](#)). A review of the developments around SMOTE can be found in [Fernández et al. \(2018\)](#).

However, neither under—nor oversampling can immediately be applied for selection bias correction since the probability sample does not always come from simple random sampling. Also, after under- or oversampling, the sample size is changed and the calculation of the correction weights should also be adjusted accordingly. In Section 2, we will start by introducing a selection bias correction framework. Since we aim for a scenario where the nonprobability sample is larger than the probability sample, the method proposed by [Liu et al. \(2022\)](#) is considered. Algorithms for under—and oversampling and a mixture of both are followed. Three evaluation studies from simulated data, synthetic data, and a real data set are presented in Section 3 to reflect the performance of the algorithms under different scenarios. Section 4 ends this paper with a short discussion and some conclusions.

2 Methods

2.1 LSW method for selection bias correction

We start by presenting the method for correcting for selection in a nonprobability sample proposed by [Liu et al. \(2022\)](#), which we will refer to as the LSW method (for details of this method, see [Liu et al., 2022](#)).

We consider a nonprobability sample (NP) with size n_{NP} from a finite population (U) indexed by $i \in \{1, 2, \dots, N\}$, with N the population size. In NP , y and $\mathbf{x}$ are available where y is a target variable and $\mathbf{x}$ is a set of auxiliary variables. For simplicity, here we consider only a single target variable y , but this can easily be generalized to more target variables. The target parameter of interest is the population mean of y , i.e. $\mu = N^{-1} \sum_U y_i$.

Along with the nonprobability sample, a reference probability sample (P) with size n_P from the same population is available with the same set of auxiliary variables $\mathbf{x}$ and design weights $d_i = 1/\mathbb{P}(i \in P)$, but not necessarily with y . Some assumptions are considered:

1. For all the units in the population, $0 < \mathbb{P}(i \in NP) < 1$ and $0 < \mathbb{P}(i \in P) < 1$;
2. $\mathbb{P}(i \in NP)$ is fully governed by $\mathbf{x}$;
3. $\mathbb{P}(i \in P)$ is available or can be calculated for $i \in P$ and $i \in NP$;
4. The overlapping units between NP and P can be identified.

Assumptions 1 and 2 are common assumptions in the selection bias correction literature to allow for valid inference. Assumptions 3 and 4 are especially for the LSW method. One way to achieve Assumption 3 is to have the design variable(s) of P as a subset of $\mathbf{x}$. If Assumption 4 does not hold, the pseudo-weighting method from [Elliott and Valliant \(2017\)](#) may be applied, although the efficiency of the resulting estimator is then less than the LSW estimator ([Liu et al., 2022](#)).

In order to estimate propensities for units to be included in NP , we fit a propensity model $f(\mathbf{x})$. This model is fitted with the non-overlapping part of the two samples (B). Within B , we assign $z_i = 1$ if $i \in NP \setminus P$, and $z_i = 0$ if $i \in P \setminus NP$ as the dependent variable. The model $f(\mathbf{x})$ can be any probability estimation method, for example, logistic regression or a classification tree. After fitting $f(\mathbf{x})$, the odds for $i \in NP$

$$O_i = \frac{\mathbb{P}(z_i = 1 | i \in B, \mathbf{x}_i)}{\mathbb{P}(z_i = 0 | i \in B, \mathbf{x}_i)} \quad (1)$$

are calculated given $f(\mathbf{x})$. The final weights, i.e. the reciprocals of the inclusion propensities, are calculated as

$$w_i = \frac{1}{\mathbb{P}(i \in NP | \mathbf{x}_i)} = 1 + \frac{d_i - 1}{O_i} \quad (2)$$

for $i \in NP$. The population mean is estimated by

$$\hat{y} = \sum_{NP} w_i y_i \left(\sum_{NP} w_i \right)^{-1}.$$

[Scholtus et al. \(2025\)](#) showed that, under some technical conditions, $\hat{y}$ is a consistent estimator of μ if the above four assumptions are satisfied. Here we focus on the scenario where the inclusion in the nonprobability sample is independent of the inclusion in the probability sample, which might not always be the case as discussed in [Liu et al. \(2022\)](#).

2.2 Balancing the two samples

When fitting a propensity model $f(\mathbf{x})$ on a nonprobability sample and a probability sample, the model may be sensitive to the imbalances between the two samples since we often have a large nonprobability sample and a small probability sample. To deal with imbalances, either under- or oversampling can be used. After fitting $f(\mathbf{x})$ with under- or over-sampled samples, the weights are then constructed for the whole NP for inference.

In undersampling, a subset is selected from the majority group by simple random sampling without replacement. Therefore, one will end up with a smaller sample size than the original data. Alternatively, one can oversample the minority group by either simply adding copies of the original data of the minority group, or creating data points that are similar to the units in the minority group.

After under- or oversampling, the calculation in (2) needs to be adjusted. Here we introduce further notation. We denote the set of units $i \in B$ with $z_i = 0$ by B_0 and the set of units $i \in B$ with $z_i = 1$ by B_1 . The size of B_0 is $n_{B0} = \sum_B (1 - z_i)$, the size of B_1 is $n_{B1} = \sum_B z_i$, and $n_{B0} + n_{B1} = n_B$. We limit our discussion to the scenario where $0 < n_{B0} < n_{B1}$. Suppose we aim to have a sample where the (original) majority group has a proportion of F . One choice of F , for example, may be 0.5 to have a fully balanced

sample. We can achieve $F = 0.5$ by undersampling the B_1 set to obtain a subset B_1^- of size $n_{B_1^-} = n_{B_0}$, or by oversampling the B_0 set to acquire a larger set B_0^+ of size $n_{B_0^+} = n_{B_1}$. Assuming all the units have the same probability of being undersampled, respectively oversampled, the calculation of weights in (2) is completed by $O_i = O_i^F / p$, where O_i^F is the odds calculated from the balanced sample and a fixed term

$$p = \frac{F}{(1-F)} \frac{n_{B_0}}{n_{B_1}}. \quad (3)$$

Finally,

$$w_i = 1 + \frac{(d_i - 1)p}{O_i^F}. \quad (4)$$

The detailed derivation of (4) can be found in the Appendix. For undersampling, p is aligned with the target sampling fraction of the units in B_1 . Algorithm 1 presents the process of undersampling.

In oversampling, studies have shown that simply adding copies of units in the minority group contributes little to model performance (Blagus & Lusa, 2013). Therefore, here we introduce SMOTE which creates a set of synthetic data points from the minority group. The general idea is that SMOTE creates synthetic data points by combining data points with their neighbours. The corresponding process is shown in Algorithm 2. The ratio of the majority group (F) is used to calculate the number of synthetic units (R) that need to be created. For every unit in B_0 , K nearest neighbours from other units in B_0 are found using a distance function based on $\mathbf{x}$. If the design variables of the probability sample are not in $\mathbf{x}$, the distance should be calculated from not only $\mathbf{x}$ but also the design weights. Considering weights in the distance calculation can leverage the auxiliary information to improve the selection of the neighbours. A similar concept can be found in the nonresponse literature where including the response propensity in the imputation model can improve the performance of the imputation (Andridge & Little, 2009).

From the literature on nearest neighbour imputation, we also learned that the efficiency of the imputation can be increased if $K > 1$ (Fuller, 2011). If K is small, the imputation bias is smaller while the imputation variance is larger. If K is large, the smoothness and symmetry of the density around the unit and its neighbours are crucial to guarantee reasonable synthetic data points. In SMOTE, after K neighbours ($\mathbf{r}_{i1}, \mathbf{r}_{i2}, \dots, \mathbf{r}_{iK}$ for unit i) are found, a random draw of the neighbours is performed and the drawn neighbour is assigned as $\mathbf{r}_q$. The drawing process can be conducted by giving all the K neighbours an equal chance to be drawn or letting $\mathbb{P}(\mathbf{r}_q = \mathbf{r}_{ik}) \propto \mathcal{D}(\mathbf{x}_i, \mathbf{r}_{ik})^{-1}$, where $\mathcal{D}$ denotes the distance function. That is, the closer the neighbour, the higher the chance of being drawn. Another random value α is drawn from $U(a, b)$, where $U(\cdot)$ is the uniform distribution on the interval (a, b) . We use $\alpha \sim U(0, 1)$ as in the original paper by Chawla et al. (2002). The synthetic data point is created by $\mathbf{s}_q = \mathbf{x}_i + \alpha(\mathbf{r}_q - \mathbf{x}_i)$. The process is repeated until $[R]$ synthetic units are created, where $[\cdot]$ denotes rounding into the closest integer. Here we presume the records in the sample are placed in a random order, or else the order should be shuffled before creating the synthetic sample, since the units with a smaller i have a higher chance of being replicated.

Algorithm 1. Undersampling

Require: d_i, B_0, B_1, NP, F

- 1: Calculate p with (3)
- 2: Draw a simple random sample without replacement from B_1 with sampling fraction p , and denote the drawn subsample by B_1^-
- 3: Fit propensity model $f(\mathbf{x})$ on $B_0 \cup B_1^-$, where $z_i = 1$ if $i \in B_1^-$ and $z_i = 0$ otherwise.
- 4: Estimate the odds O_i^F of the units in NP given $f(\mathbf{x})$.
- 5: Calculate the weights with $w_i = 1 + (d_i - 1)p/O_i^F$.

Algorithm 2. Oversampling by SMOTE**Require:** d_i, B_0, B_1, NP, F, K

- 1: Calculate the size of synthetic data $R = n_{B1}/F - n_B$ and round it to $[R]$
- 2: Find K nearest neighbours (r_{ik}) from B_0 for every $i \in B_0$
- 3: Assign $i \leftarrow 1$
- 4: **for** $q = 1$ to $[R]$ **do**
- 5: Draw a neighbour of i from $\{r_{i1}, r_{i2}, \dots, r_{ik}\}$ and assign as r_q
- 6: Draw $\alpha \sim U(0, 1)$
- 7: Create a synthetic data point by $s_q = x_i + \alpha(r_q - x_i)$
- 8: Assign $i \leftarrow i + 1$. If $i = n_{B0} + 1$, then $i \leftarrow 1$.
- 9: **end for**
- 10: Row combine all the s_q with B_0 to form B_0^+
- 11: Fit propensity model $f(x)$ on $B_0^+ \cup B_1$, where $z_i = 1$ if $i \in B_1$ and $z_i = 0$ otherwise.
- 12: Estimate the odds O_i^f for the units in NP given $f(x)$.
- 13: Calculate p with (3) and the weights with $w_i = 1 + (d_i - 1)p/O_i^f$.

Given the new samples generation process, SMOTE improves the propensity estimation if the generated samples follow the population distribution of $\mathbf{x}$. It is easier to achieve the similarity of the true and synthetic distributions if the population distribution of $\mathbf{x}$ is symmetric and smooth. Also, the samples are drawn by a high entropy sampling method, and the dimensionality of $\mathbf{x}$ is not too high. The new samples may then provide a stronger influence on the target objective function to decrease the bias toward the majority group (Elreedy & Atiya, 2019).

When the data is extremely imbalanced, $n_{B0} \ll n_{B1}$, the choice between under- or oversampling may be difficult. Undersampling implies that a large proportion of B_1 is discarded, and the model is fitted with a small sample size. On the other hand, in oversampling, a large amount of synthetic data is created given limited observed data. The distribution of the synthetic data may differ a lot from the true distribution of the minority group. As a trade-off, instead of merely under- or oversampling, another option is to perform both under- and oversampling at the same time. The process is shown in Algorithm 3 where the aim of the algorithm is to create a balanced sample with a sample size equal to n_B .

3 Evaluations

In this section, we evaluate the performance of the algorithms on different types of data. In the first evaluation, the data are simulated from symmetric distributions, and the number of variables is small, which is an ideal scenario for SMOTE (Blagus & Lusa, 2013). In the second evaluation, we apply the algorithms to a synthetic data set that mimics a real-world data set, reflecting a more complex sampling design and practical decision-making. And finally, a real-world data set is used to reflect a completely uncontrolled scenario.

3.1 Evaluation on simulated data

A finite population U of size $N = 30,000$ is created with $y, x_1, x_2, x_3 \sim N(\mu = 1, \sigma = 1)$, i.e. the normal distribution with mean $\mu = 1$ and standard deviation $\sigma = 1$. We are interested in estimating the population mean of the target variable y . The true population mean $\mu = N^{-1} \sum_U y_i \approx 1$. To reflect different amounts of auxiliary information researchers have, the correlation between y and every auxiliary variable x is 0.7 as more informative auxiliary information, and the correlation between y and each x is 0.3 as less informative auxiliary information. The correlation between each x is 0.3. Selective nonprobability samples are drawn with inclusion probabilities $\pi_i = \exp(\beta_0 - 2y_i) / [1 + \exp(\beta_0 - 2y_i)]$, where β_0 is a constant allowing control of the sample size of the nonprobability sample. It is worth noting that it is

Algorithm 3. A mixture of SMOTE and undersampling**Require:** d_i, B_0, B_1, NP, F, K

-
- 1: Draw a simple random sample without replacement from B_1 with size $F \times n_{B_1}$, and denote this sample by B_1^-
 - 2: Calculate the size of synthetic data $R = n_B(1 - F) - n_{B_0}$ and round it to $[R]$
 - 3: Find K nearest neighbours (r_{ik}) from B_0 for every $i \in B_0$
 - 4: Assign $i \leftarrow 1$
 - 5: **for** $q = 1$ to $[R]$ **do**
 - 6: Draw a neighbour of i from $\{r_{i1}, r_{i2}, \dots, r_{ik}\}$ and assign as r_q
 - 7: Draw $\alpha \sim U(0, 1)$
 - 8: Create a synthetic data point by $s_q = x_i + \alpha(r_q - x_i)$
 - 9: Assign $i \leftarrow i + 1$. If $i = n_{B_0} + 1$, then $i \leftarrow 1$.
 - 10: **end for**
 - 11: Row combine all the s_q with B_0 to form B_0^+
 - 12: Fit propensity model $f(x)$ on $B_0^+ \cup B_1^-$, where $z_i = 1$ if $i \in B_1^-$ and $z_i = 0$ otherwise.
 - 13: Estimate the odds O_i^f of the units in NP given $f(x)$.
 - 14: Calculate p with (3) and the weights with $w_i = 1 + (d_i - 1)p/O_i^f$.
-

non-ignorable selection, or missing not at random in a missing data context. Although this violates the Assumption 2 in Section 2, it reflects a more realistic practical scenario. The inclusion fractions of the nonprobability sample (f_{NP}) are 10, 30, 50, and 70 percent, and the inclusion fractions of the probability sample (f_P) are 1, 5, 10, 30, and 50 percent to reflect different levels of imbalances. We only look at the combinations of inclusion fractions where $f_P < f_{NP}$, which results in 14 combinations. Two sampling designs of the probability sample are considered, namely Simple Random Sampling (SRS) and Probability Proportional to Size (PPS). When the probability sample is from a PPS design, the inclusion probability is proportional to x_1 . Several approaches are conducted for comparison:

- Naive: Treat the nonprobability sample as a simple random sample.
- LSW: Apply the LSW method without any adjustment for imbalance.
- Undersampling (US): Draw a subsample of the nonprobability sample by Algorithm 1.
- Oversampling with SMOTE (OS): Create synthetic samples by Algorithm 2.
- Mix: A combination of undersampling and SMOTE by Algorithm 3.

The under- or oversampling is done in such a way that the B sample is fully balanced, that is, $F = 0.5$ so that around half of the units in the final sample are units with $z_i = 1$ and the other half with $z_i = 0$. The number of nearest neighbours is $K = 5$ for all the algorithms applying SMOTE and the R package *smotefamily* is used. The propensity model is a logistic regression fitted with all the auxiliary variables. The performance of each approach is evaluated with estimated Relative Bias $RB(\hat{y}) = M^{-1} \sum_{m=1}^M (\hat{y}_m - \mu)/\mu$ and mean squared error $MSE(\hat{y}) = M^{-1} \sum_{m=1}^M (\hat{y}_m - \mu)^2$ where m is the index of simulation runs with $m = 1, 2, \dots, M$ and $M = 1,000$. RB and MSE are both often-used performance indicators for bias correction, where RB reflects the percentage of improvement and MSE reflects the combined effect of bias and variation.

The results from SRS samples are in Table 1, and the results of PPS samples are in Table 2. In general, SMOTE performs well across different scenarios. Undersampling does contribute to decreasing bias in some cases, but it does not compensate for the large variation. Although the mixture of under- and oversampling does not perform the best, it is still better than doing nothing to balance the samples (as in LSW) most of the time. The performance in general is not affected much with respect to the level of the imbalances. Only in the most imbalanced cases, that is, $f_P = 1$, $f_{NP} = 70$, with informative auxiliary variables, LSW shows the lowest MSE while other methods show improvement of relative bias. This reflects that the corrected bias is not sufficient to compensate for the introduced variance.

Table 1. MSE and relative bias for simulated SRS samples

%	f_P		f_{NP}	100*MSE				RB (%)					
				Naive	LSW	OS	US	Mix	Naive	LSW	OS	US	Mix
1	1	10	163.80	3.78	3.44	5.02	3.66	-127.88	-15.49	-10.01	-14.49	-10.49	
5	5	10	163.86	4.07	3.85	4.29	4.18	-127.91	-17.02	-15.89	-17.10	-17.09	
1	1	30	76.59	1.65	1.29	2.55	1.31	-87.44	-11.79	-6.00	-9.52	-6.11	
5	5	30	76.49	1.70	1.43	1.98	1.51	-87.39	-12.31	-10.62	-12.62	-11.02	
10	10	30	76.53	1.76	1.62	1.94	1.69	-87.41	-12.69	-11.95	-13.04	-12.15	
1	1	50	36.93	0.78	0.64	1.64	0.64	-60.72	-8.06	-2.41	-5.92	-2.44	
5	5	50	36.94	0.76	0.55	0.92	0.58	-60.73	-8.38	-6.45	-8.05	-6.60	
10	10	50	36.92	0.78	0.62	0.87	0.65	-60.72	-8.57	-7.34	-8.58	-7.51	
30	30	50	36.94	0.79	0.73	0.80	0.79	-60.73	-8.63	-8.22	-8.62	-8.62	
1	1	70	14.22	0.31	0.64	1.63	0.65	-37.68	-4.67	1.87	0.15	1.87	
5	5	70	14.21	0.26	0.15	0.35	0.15	-37.67	-4.84	-2.71	-3.99	-2.75	
10	10	70	14.21	0.25	0.15	0.28	0.16	-37.67	-4.86	-3.33	-4.28	-3.41	
30	30	70	14.20	0.25	0.19	0.25	0.20	-37.66	-4.87	-4.14	-4.66	-4.23	
50	50	70	14.21	0.25	0.22	0.24	0.25	-37.66	-4.88	-4.54	-4.73	-4.85	
1	1	10	163.05	125.48	124.90	126.11	125.07	-127.36	-111.70	-111.45	-111.97	-111.52	
5	5	10	163.09	126.02	126.02	126.27	126.10	-127.37	-111.96	-111.95	-112.07	-111.99	
1	1	30	75.76	59.29	58.69	59.07	58.69	-86.81	-76.78	-76.39	-76.62	-76.39	
5	5	30	75.83	59.49	59.40	59.63	59.44	-86.85	-76.92	-76.87	-77.01	-76.89	
10	10	30	75.80	59.55	59.53	59.68	59.55	-86.84	-76.96	-76.95	-77.05	-76.97	
1	1	50	36.34	28.27	27.83	28.15	27.83	-60.12	-53.01	-52.60	-52.89	-52.60	
5	5	50	36.31	28.27	28.11	28.25	28.12	-60.10	-53.03	-52.87	-53.01	-52.88	
10	10	50	36.32	28.29	28.21	28.31	28.22	-60.11	-53.05	-52.97	-53.07	-52.98	
30	30	50	36.31	28.28	28.26	28.29	28.29	-60.10	-53.04	-53.02	-53.05	-53.05	
1	1	70	13.91	10.67	10.37	10.42	10.37	-37.19	-32.56	-32.09	-32.15	-32.09	
5	5	70	13.91	10.61	10.48	10.58	10.48	-37.20	-32.48	-32.28	-32.43	-32.29	
10	10	70	13.91	10.61	10.53	10.58	10.53	-37.20	-32.49	-32.36	-32.44	-32.37	
30	30	70	13.90	10.60	10.56	10.59	10.57	-37.19	-32.47	-32.42	-32.45	-32.42	
50	50	70	13.91	10.60	10.58	10.59	10.60	-37.20	-32.47	-32.44	-32.45	-32.47	

Note. The upper block corresponds to the condition with informative auxiliary variables, and the lower block with less informative auxiliary variables. In each row, the smallest (absolute) value is printed in bold.

Table 2. MSE and relative bias for simulated PPS samples

% f_P			100*MSE				RB (%)					
f_{NP}			Naive	LSW	OS	US	Mix	Naive	LSW	OS	US	Mix
1	1	10	163.86	3.85	3.54	5.20	3.68	-127.90	-15.70	-8.60	-13.71	-9.38
5	5	10	163.80	4.16	3.92	4.37	4.19	-127.88	-15.55	-13.93	-15.04	-15.38
1	1	30	76.52	1.59	1.30	2.63	1.33	-87.41	-11.48	-4.80	-9.23	-4.96
5	5	30	76.56	1.60	1.22	1.68	1.28	-87.43	-11.89	-9.61	-11.25	-9.84
10	10	30	76.58	1.60	1.37	1.70	1.43	-87.44	-11.90	-10.69	-11.84	-10.89
1	1	50	36.94	0.78	0.64	1.61	0.66	-60.73	-8.14	-2.05	-4.92	-2.09
5	5	50	36.95	0.75	0.49	0.85	0.51	-60.74	-8.34	-5.94	-7.60	-6.09
10	10	50	36.92	0.74	0.54	0.74	0.58	-60.71	-8.29	-6.74	-7.80	-7.00
30	30	50	36.95	0.75	0.67	0.76	0.75	-60.74	-8.42	-7.85	-8.29	-8.37
1	1	70	14.21	0.31	0.55	1.37	0.55	-37.66	-4.81	1.96	-0.05	1.89
5	5	70	14.21	0.25	0.12	0.31	0.13	-37.67	-4.76	-2.33	-3.60	-2.41
10	10	70	14.22	0.24	0.14	0.25	0.15	-37.67	-4.78	-3.08	-4.02	-3.19
30	30	70	14.21	0.24	0.18	0.22	0.20	-37.67	-4.76	-3.95	-4.32	-4.19
50	50	70	14.22	0.28	0.25	0.28	0.28	-37.68	-5.18	-4.80	-5.05	-5.12
1	1	10	163.22	125.60	124.55	125.74	124.65	-127.42	-111.76	-111.28	-111.80	-111.33
5	5	10	163.05	125.86	125.68	126.04	125.96	-127.35	-111.88	-111.80	-111.96	-111.93
1	1	30	75.82	59.61	58.67	59.50	58.68	-86.84	-76.99	-76.38	-76.90	-76.39
5	5	30	75.79	59.48	59.19	59.54	59.23	-86.83	-76.92	-76.73	-76.96	-76.76
10	10	30	75.79	59.52	59.38	59.58	59.42	-86.83	-76.94	-76.85	-76.98	-76.88
1	1	50	36.31	28.43	27.73	28.13	27.74	-60.10	-53.17	-52.51	-52.86	-52.51
5	5	50	36.32	28.36	28.06	28.28	28.07	-60.11	-53.11	-52.83	-53.03	-52.84
10	10	50	36.31	28.35	28.15	28.28	28.17	-60.10	-53.10	-52.91	-53.03	-52.93
30	30	50	36.34	28.33	28.27	28.31	28.33	-60.12	-53.09	-53.03	-53.06	-53.08
1	1	70	13.91	10.69	10.27	10.40	10.27	-37.19	-32.60	-31.94	-32.10	-31.94
5	5	70	13.91	10.67	10.47	10.59	10.48	-37.19	-32.58	-32.27	-32.44	-32.28
10	10	70	13.91	10.66	10.51	10.58	10.51	-37.20	-32.57	-32.33	-32.44	-32.34
30	30	70	13.91	10.63	10.55	10.58	10.56	-37.19	-32.51	-32.39	-32.44	-32.40
50	50	70	13.90	10.63	10.61	10.63	10.62	-37.19	-32.51	-32.48	-32.51	-32.51

Note. The upper block is the result of informative auxiliary variables, and the lower block is the result of less informative auxiliary variables. The smallest (absolute) value of each row is bold.

3.2 Evaluation on synthetic data

In the second evaluation, we use a synthetic business population consisting of 900,000 records; see [Ang et al. \(2024\)](#) for the location of the data set, along with information on how it was created. The data set aims to resemble the real-world population of employing businesses in Australia in terms of the distribution of businesses across industry groupings, geographic locations, and business size. Data for the population was generated by using a combination of published business outputs from the Australian Bureau of Statistics (ABS), as well as business survey information and employee tax data sourced from the Business Longitudinal Analysis Data Environment (BLADE) in the ABS DataLab. A goal of the data generation process was to ensure the characteristics of the variables created would mimic their real-world counterparts. Variables available in the data set include:

- *earnings*—Total weekly wages and salaries reported by the business
- *rep.emp*—Total number of employees reported by the business
- *ovt*—Total overtime paid to employees by the business
- *frame.emp*—An out-of-date value for the total number of employees in the business
- *indgrp*—Industry class for the business
- *size2*—Business size grouping based on the *frame.emp* variable
- *state*—Broad geographic identifier for the business

The target variables are *ovt* and *earnings*, which descriptive statistics are shown in [Table 3](#). Both target variables are extremely skewed. The mean of *ovt* is 519 and 76% of the data is zero. For *earnings*, the mean is 20,090 and 9.3% of the data is zero. The nonprobability sample is drawn with inclusion probabilities proportional to *pi.mar*, where *pi.mar* was generated using a two-stage process. At the first stage, an initial probability $pi.mar_1$ was produced, where

$$pi.mar_1 = \frac{\exp(\phi_0 + \phi_1 x)}{1 + \exp(\phi_0 + \phi_1 x)}$$

with $(\phi_0, \phi_1) = (0.09, 0.009)$, and x is the *frame.emp* variable. At the second stage, the $pi.mar_1$ values were adjusted downwards by a pre-specified factor in some *indgrp* classes, to simulate a reduced likelihood of being present on the big data set for those industries. The resulting probabilities were the final *pi.mar* values. The modelled inclusion probabilities result in units with smaller frame employment having a lower chance of being included in the nonprobability sample.

The same approaches are compared as in [Section 3.1](#). Since the data set is a combination of continuous data and categorical data, the package *themis* is used for SMOTE where the categorical variable is created by the majority category of the neighbours.

3.2.1 SRS sample

Again we look at the scenario where the probability sample is a simple random sample. The inclusion fractions of the probability sample are 0.1% and 0.5%, and the inclusion fractions of the nonprobability sample are 1% and 3%. Both samples are drawn 300 times. The propensity model is fitted with

Table 3. Descriptive statistics of the target variables

	ovt	earnings
Minimum	0	0
1st quartile	0	3,819
Median	0	9,459
Mean	519	20,090
3rd quartile	0	16,619
Maximum	1,259,194	8,262,828

Table 4. MSE and relative bias for synthetic SRS samples

	%		MSE				
	f_P	f_{NP}	Naive	LSW	OS	US	Mix
ovt	0.1	1	231.328	51.221	28.869	3679.292	27.036
	0.5	1	247.183	38.716	29.056	40.843	32.122
	0.1	3	211.640	36.530	17.763	1499.057	17.893
	0.5	3	215.571	27.035	13.617	31.120	15.737
earnings	0.1	1	308.846	45.490	21.734	538.798	20.599
	0.5	1	307.609	33.824	21.470	35.251	26.204
	0.1	3	296.198	46.346	21.396	657.055	20.916
	0.5	3	291.531	30.083	12.718	35.645	15.240
RB (%)							
ovt	0.1	1	26.434	7.704	-2.642	19.832	-1.677
	0.5	1	27.039	7.876	5.784	7.866	6.365
	0.1	3	27.024	8.081	-3.174	16.814	-2.612
	0.5	3	27.152	8.067	4.539	8.362	5.116
earnings	0.1	1	27.218	7.838	-2.964	14.207	-1.908
	0.5	1	27.115	8.055	5.908	7.936	6.583
	0.1	3	26.947	8.191	-3.392	14.083	-2.887
	0.5	3	26.701	7.836	4.339	8.047	4.913

Note. The smallest (absolute) value of each row is bold.

auxiliary variables *size2* and *indgrp* with logistic regression. Note that we fit the propensity model with the categorized version of *frame.emp* (*size2*) instead of the original *frame.emp* since *frame.emp* contains a lot of zeros and results in unstable estimated models.

The results are shown in Table 4. The presented MSE of *ovt* is 10^{-2} of the original MSE, and the presented MSE of *earnings* is 10^{-5} times the original MSE. Although the distributions of the target variables are extremely skewed as shown in Table 3, both oversampling and the mixture of under- and oversampling improve the estimates regarding MSE and relative bias. Undersampling, however, is worse than no adjustment in most cases.

3.2.2 Stratified SRS sample

A stratified SRS sample is considered to reflect a practical scenario. A realistic business survey sample design was conducted by first assigning the population data set into strata based on State (*state*), Industry class (*indgrp*), and Size (*size2*). The size categories were: 0–4 employees, 5–19 employees, 20–299 employees, and 300+ employees. The Bethel-Chromy algorithm (Bethel, 1989; Chromy, 1987) was used to produce an optimal allocation of sample to strata, to meet the following accuracy constraints for the *earnings* variable:

- Relative Standard Error (RSE) of 1.5% at the National level
- RSE of 5% for each Industry class
- RSE of 5% for each State

where RSE is calculated as the standard error of *earnings* over the estimated value of *earnings*. A minimum sample size of 6 was applied for each sampled stratum, while the large units (that is, those in the 300+ Size strata) were fully sampled with a sampling fraction of 1. Simple random sampling is applied in those strata with $\mathbb{P}(i \in P) \neq 1$. In total, the sample size of the stratified SRS sample is 7,715 ($f_p = 0.86\%$), with 3,259 units having $\mathbb{P}(i \in P) = 1$.

Being fully sampled, the large units in the population have $\mathbb{P}(i \in P) = 1$, which is often seen in the business survey scenario but it violates the first assumption as discussed in Section 2. Therefore, we separate the population into two sets. Set CE contains units with $\mathbb{P}(i \in P) = 1$, and set $U \setminus CE$ contains units with $\mathbb{P}(i \in P) < 1$. In the set $U \setminus CE$, all the estimation steps are the same as before and the weight is denoted as $W_{U \setminus CE}$. The propensity model is fitted with *size2*, *indgrp*, and *state*. In CE , since only the units in the nonprobability sample have a value of y , another set of weights is constructed. A propensity model is fitted in CE with $z_i = 1$ if $i \in NP$ and $z_i = 0$ otherwise. The weights are then

$$W_{CE} = \frac{1}{\mathbb{P}(i \in NP | i \in CE, \mathbf{x}_i)}.$$

The final estimate of μ is

$$\hat{y} = f_{CE} \frac{\sum_{i \in NP \cap CE} y_i W_{CE,i}}{\sum_{i \in NP \cap CE} W_{CE,i}} + (1 - f_{CE}) \frac{\sum_{i \in NP \cap (U \setminus CE)} y_i W_{U \setminus CE,i}}{\sum_{i \in NP \cap (U \setminus CE)} W_{U \setminus CE,i}},$$

where f_{CE} is the proportion of the CE set in the population. The nonprobability samples are drawn as in the SRS sample cases, with inclusion fractions $f_{NP} = \{3, 5, 7, 10, 30\}$ percent.

The results are shown in Table 5. The presented MSE is again the original MSE times 10^{-2} for *ovt* and times 10^{-5} for *earnings*. Balancing two samples with either under- or oversampling does not improve the performance in many cases when the probability sample is from a stratified SRS. The only improvement we found is that after undersampling, *earning* has a smaller relative bias. This result reflects how the design of the probability sample affects the performance of under- or oversampling.

Table 5. MSE and relative bias for synthetic stratified SRS samples

	%	f_{NP}	MSE				
			Naive	LSW	OS	US	Mix
ovt	3		217.845	4.459	4.773	4.903	4.844
	5		206.626	3.305	3.809	4.209	3.880
	7		203.416	1.968	2.462	2.615	2.481
	10		195.689	1.145	1.596	1.708	1.619
	30		198.902	0.457	2.585	2.567	2.589
earnings	3		299.790	1.019	0.973	0.994	0.939
	5		293.386	0.740	0.942	0.977	0.978
	7		293.815	0.539	1.004	0.848	1.007
	10		291.984	0.371	1.041	0.809	1.033
	30		291.497	0.592	3.319	2.468	3.314
RB (%)							
ovt	3		27.471	0.615	1.014	0.223	0.950
	5		27.014	0.486	1.173	0.573	1.138
	7		27.082	0.314	1.364	0.550	1.333
	10		26.694	-0.043	1.259	0.480	1.249
	30		27.106	0.550	2.793	2.277	2.792
earnings	3		27.091	1.073	1.045	0.227	0.940
	5		26.883	0.991	1.199	0.450	1.162
	7		26.927	0.910	1.409	0.546	1.378
	10		26.861	0.728	1.453	0.693	1.434
	30		26.868	1.028	2.742	2.096	2.738

Note. The smallest (absolute) value of each row is bold.

Table 7. Selected quantiles for turnover and wages

	Quantile				
	1%	25%	50%	75%	99%
Turnover	\$0	\$59,408	\$147,527	\$403,741	\$10,101,149
Wages	\$0	\$15,150	\$39,108	\$102,078	\$2,074,381

Note. Derived population counts created using BLADE indicative items and BAS data from Quarter 1 2022–2023, as well as PAYG data from 2022–2023 ([Business Longitudinal Analysis Data Environment \(BLADE\), 2022](#)) and BAS data from Quarter 1 2022–2023, accessed through the ABS DataLab December 2024.

Table 8. MSE and relative bias for SRS samples from real data

	%		MSE ($\times 10^{-6}$)				
	f_P	f_{NP}	Naive	LSW	OS	US	Mix
turnover	0.01	0.1	908.57	87.23	90.00	93.88	90.99
	0.02	0.1	791.47	76.00	77.07	77.94	77.36
	0.01	0.2	698.54	30.66	32.25	33.71	32.23
	0.02	0.2	690.66	31.33	31.39	31.28	31.60
turnover			RB (%)				
	f_P	f_{NP}	Naive	LSW	OS	US	Mix
	0.01	0.1	30.31	−0.9	−0.29	−0.49	−0.34
	0.02	0.1	28.29	−2.2	−1.41	−2.03	−1.45
	0.01	0.2	29.22	−0.44	−0.4	−0.27	−0.42
	0.02	0.2	29.07	−0.63	−0.2	−0.36	−0.25

Note. The smallest (absolute) value of each row is bold.

the results. The unmodified LSW approach results in the lowest MSE values, while oversampling results in the lowest RB values in 3 of the 4 sampling scenarios. The OS, US and mixture of under- and oversampling perform relatively similarly in terms of MSE, with the OS method generally coming closest to the LSW results.

3.3.2 Stratified SRS sample

A realistic business survey sample design was conducted by first assigning the population data set into strata based on State, Industry class, and Size. The size categories were: 0–4 employees, 5–19 employees, 20–299 employees, and 300+ employees. The Bethel-Chromy algorithm ([Bethel, 1989; Chromy, 1987](#)) was used to produce an optimal allocation of sample to strata, to meet the following accuracy constraints for Wages:

- RSE of 3% at the National level
- RSE of 5% for each Industry class

A minimum sample size of 6 was applied for each sampled stratum, while the large units belonging in the 300+ size strata were fully sampled. In total, the sample size of the stratified SRS sample was 14,929. To account for the fact that there is a fraction of the sample with sampling fraction of 1, we applied the same approach described in Section 3.2.2.

Table 9 provides the results for this sample design scenario. The results are mixed, with the under-sampling providing the lowest MSE when the size of the non-probability sample is smaller, while the mixture and oversampling approaches perform better for the larger non-probability sample sizes. In terms

Table 9. MSE and relative bias for stratified SRS samples

	%		MSE ($\times 10^{-6}$)				
	f_P	f_{NP}	Naive	LSW	OS	US	Mix
turnover	0.02	0.05	1,005	1,835	1,469	990	1,305
	0.02	0.10	812	1,193	747	541	699
	0.02	0.20	670	416	275	331	274
	0.02	0.30	618	154	171	280	174
RB (%)							
turnover	0.02	0.05	27.81	-7.14	-10.98	-17.88	-12.8
	0.02	0.10	28.87	0.67	-7.44	-16.13	-8.68
	0.02	0.20	28.58	-0.66	-10.35	-18.03	-10.89
	0.02	0.30	28.36	-2.84	-11.94	-18.44	-12.2

Note. The smallest (absolute) value of each row is bold.

of RB, the oversampling, undersampling and mixture approaches all have some residual selection bias. The LSW approach yields the lowest RB values of the four approaches examined, although for the smallest and largest non-probability samples it also exhibits some residual bias.

4 Conclusion and discussion

When correcting selection bias with a two-sample setup, an imbalanced combined sample is often seen and may hinder the propensity estimation. In this research, we evaluate algorithms that aim to deal with imbalance in the prediction problem in a selection bias correction framework. Three evaluations are presented: from a fully simulated population, a synthetic population mimicking a real data set, to an actual real data set. Three different evaluations come with different distributions, and two types of sampling designs of the probability sample are considered. In general, across the three evaluations, we found that the inclusion fractions of the two samples do not play an important role in the performance. Among all the under- and oversampling methods, oversampling by SMOTE shows the potential to decrease the MSE and relative bias. However, in some cases, SMOTE does not perform better than not having any adjustment. In the second evaluation (synthetic data), the design of the probability sample seems to play an important role, while this effect does not replicate in the real data cases. An evaluation method is needed to assist researchers in determining whether a balancing method is needed under the given data set.

This research inevitably can only investigate a limited number of modelling options. Variants of SMOTE may be examined for future studies. For example, different distance functions for choosing the nearest neighbours, different ways to draw the nearest neighbours, and different ways to combine the neighbours and the sample. Also, although we illustrate how to remedy the imbalance problem with the LSW method, other selection bias correction methods (e.g. [Chen et al., 2020](#)) may also benefit from the balanced samples as long as the calculation of the correction weights is adjusted accordingly.

Besides the point estimate, the variance of the estimates is often of interest. In the original study of [Liu et al. \(2022\)](#), nonparametric variance estimations are proposed, and later analytical formulas can be found in [Scholtus et al. \(2025\)](#). Further research is needed to incorporate the variation from under- and oversampling in the variance estimate. Also, logistic regression may not always be the best model for propensity estimation, especially when the inclusion mechanism is non-linear. Given the nature of the LSW method, researchers may replace logistic regression with other promising models, for example, machine learning algorithms. However, this was investigated by [Ciawandy \(2025\)](#) and it shows that logistic regression is a rather stable method. Further investigation may be needed to determine whether the discussed algorithms improve the results under different models.

Conflicts of interest

The authors declare no competing interests.

Funding

There is no funding related to this research.

Data availability

The synthetic data set can be found at <https://doi.org/10.5281/zenodo.11095755>, and the code can be found at <https://github.com/cheerup731/lmbalanced>.

ABS DataLab disclaimer

The data used to create the derived population for the business turnover application in this paper was supplied to the ABS under the Taxation Administration Act 1953, A New Tax System (Australian Business Number) Act 1999, Australian Border Force Act 2015, Social Security (Administration) Act 1999, A New Tax System (Family Assistance) (Administration) Act 1999, Paid Parental Leave Act 2010 and/or the Student Assistance Act 1973. Such data may only used for the purpose of administering the Census and Statistics Act 1905 or performance of functions of the ABS as set out in Section 6 of the Australian Bureau of Statistics Act 1975. No individual information collected under the Census and Statistics Act 1905 is provided back to custodians for administrative or regulatory purposes. Any discussion of data limitations or weaknesses is in the context of using the data for statistical purposes and is not related to the ability of the data to support the Australian Taxation Office, Australian Business Register, Department of Social Services and/or Department of Home Affairs' core operational requirements.

Legislative requirements to ensure privacy and secrecy of these data have been followed. For access to PLIDA and/or BLADE data under Section 16A of the ABS Act 1975 or enabled by Section 15 of the Census and Statistics (Information Release and Access) Determination 2018, source data are de-identified and so data about specific individuals has not been viewed in conducting this analysis. In accordance with the Census and Statistics Act 1905, results have been treated where necessary to ensure that they are not likely to enable identification of a particular person or organization.

Appendix

Here we present the derivation of (4).

Undersampling

For the proposed undersampling approach (Algorithm 1), the propensity model is estimated on $B^- = B_0 \cup B_1^-$. That is to say, instead of $O_i(B)$ from (1) we model

$$O_i(B^-) = \frac{\mathbb{P}(z_i = 1 \mid i \in B^-, \mathbf{x}_i)}{\mathbb{P}(z_i = 0 \mid i \in B^-, \mathbf{x}_i)}.$$

To find the correct pseudo weights based on (2), we can work out the relationship between $O_i(B^-)$ and $O_i(B)$.

Define the subsampling fraction $p^- = n_{B_0}F/n_{B_1}(1 - F)$. We obtain:

$$\begin{aligned} \frac{\mathbb{P}(z_i = 1 \mid i \in B^-, \mathbf{x}_i)}{\mathbb{P}(z_i = 0 \mid i \in B^-, \mathbf{x}_i)} &= \frac{\mathbb{P}(z_i = 1, i \in B^- \mid \mathbf{x}_i)}{\mathbb{P}(z_i = 0, i \in B^- \mid \mathbf{x}_i)} \\ &= \frac{\mathbb{P}(z_i = 1, i \in B^-, i \in B \mid \mathbf{x}_i)}{\mathbb{P}(z_i = 0, i \in B^-, i \in B \mid \mathbf{x}_i)} \\ &= \frac{\mathbb{P}(z_i = 1, i \in B \mid \mathbf{x}_i) \cdot \mathbb{P}(i \in B^- \mid z_i = 1, i \in B, \mathbf{x}_i)}{\mathbb{P}(z_i = 0, i \in B \mid \mathbf{x}_i) \cdot \mathbb{P}(i \in B^- \mid z_i = 0, i \in B, \mathbf{x}_i)} \\ &= \frac{\mathbb{P}(z_i = 1, i \in B \mid \mathbf{x}_i) \cdot p^-}{\mathbb{P}(z_i = 0, i \in B \mid \mathbf{x}_i)} \\ &= \frac{\mathbb{P}(z_i = 1 \mid i \in B, \mathbf{x}_i)}{\mathbb{P}(z_i = 0 \mid i \in B, \mathbf{x}_i)} p^-. \end{aligned}$$

In the second line, it was used that $B^- \subseteq B$. In the fourth line, it was used that

$$\mathbb{P}(i \in B^- \mid z_i = 1, i \in B, \mathbf{x}_i) = \mathbb{P}(i \in B^- \mid z_i = 1, i \in B) = p^-,$$

since the subsample is obtained by simple random sampling without replacement from B_1 . In the same line, it was also used that $\mathbb{P}(i \in B^- \mid z_i = 0, i \in B, \mathbf{x}_i) = 1$.

We conclude that $O_i(B^-) = O_i(B)p^-$ or, equivalently, $O_i(B) = O_i(B^-)/p^-$. Hence, with this undersampling approach we can compute the pseudo weights according to (2) as follows:

$$w_i = \frac{1}{\mathbb{P}(i \in NP \mid \mathbf{x}_i)} = 1 + \frac{d_i - 1}{O_i(B^-)} p^-. \quad (5)$$

Oversampling by SMOTE

For the proposed oversampling approach based on the original SMOTE method (Algorithm 2), the propensity model is estimated on $B^+ = B_0^+ \cup B_1$ to predict

$$O_i(B^+) = \frac{\mathbb{P}(z_i = 1 \mid i \in B^+, \mathbf{x}_i)}{\mathbb{P}(z_i = 0 \mid i \in B^+, \mathbf{x}_i)}.$$

This time, we work out the relationship between $O_i(B^+)$ and $O_i(B)$.

If we ignore the effect of rounding in Algorithm 2 and assume that the records in B_0 are placed in a random order before running the algorithm, it holds that

$$\mathbb{P}(i \in B \mid i \in B_0^+, \mathbf{x}_i) = \mathbb{P}(i \in B \mid i \in B_0^+) = \frac{|B \cap B_0^+|}{|B_0^+|} = \frac{|B_0|}{|B_0^+|} \equiv p^+.$$

Furthermore,

$$|B_0^+| = |B_0| + R = n_{B_0} + \frac{n_{B_1}}{F} - (n_{B_0} + n_{B_1}) = n_{B_1} \frac{1 - F}{F}.$$

Hence, the probability p^+ can be expressed as $p^+ = n_{B_0}F/n_{B_1}(1 - F) (= p^-)$.

We can now apply a similar derivation as in the case of undersampling, but here it is convenient to start with $O_i(B)$ rather than $O_i(B^+)$. (Note that this time $B \subseteq B^+$ whereas previously we had $B^- \subseteq B$.) We obtain:

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